Cambridge (CIE) IGCSE Maths: Extended

Your notes

Further Graphs & Tangents

Contents

Types of Graphs

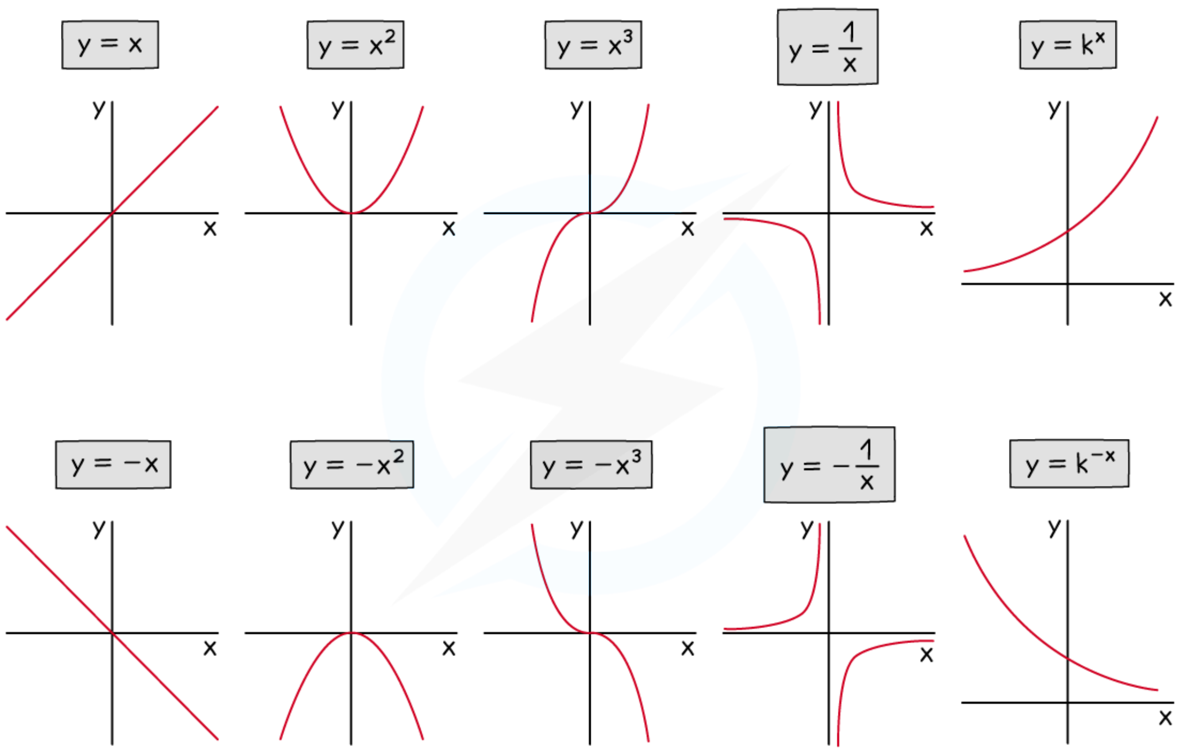
Your notes

Types of Graphs

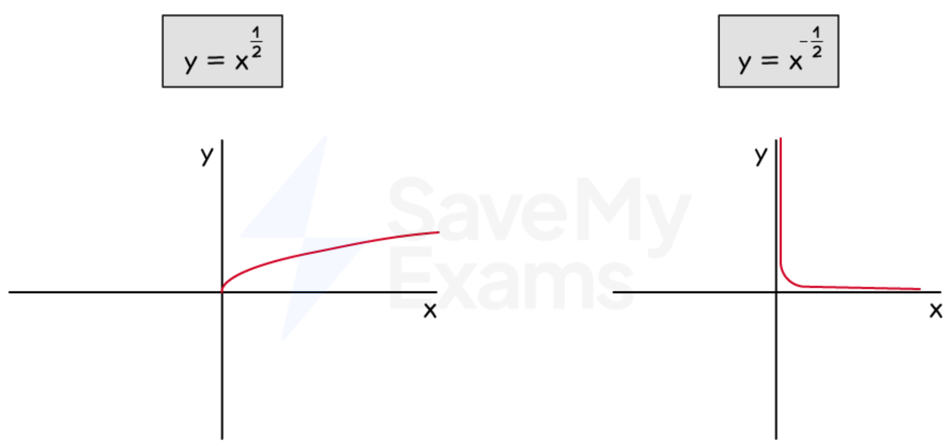
What types of graphs do I need to know?

- You need to be able to **recognise, sketch, and interpret** the following types of graph:
  - **Linear** ($y = \pm x$)
    - $y = mx + c$ or $ax + by = c$
  - **Quadratic** ($y = \pm x^2$)
    - $y = ax^2 + bx + c$
  - **Cubic** ($y = \pm x^3$)
    - $y = ax^3 + b$ or $y = ax^3 + bx^2 + cx$
  - **Reciprocal** ($y = \pm \frac{1}{x}$)
    - $y = \frac{a}{x} + b$
  - **Exponential** ($y = k^{\pm x}$)
    - $y = ak^x + b$

Your notes



- You may also need to recognise the graphs $y = x^{\frac{1}{2}}$ and $y = x^{-\frac{1}{2}}$
- These are equivalent to $y = \sqrt{x}$ and $y = \dfrac{1}{\sqrt{x}}$

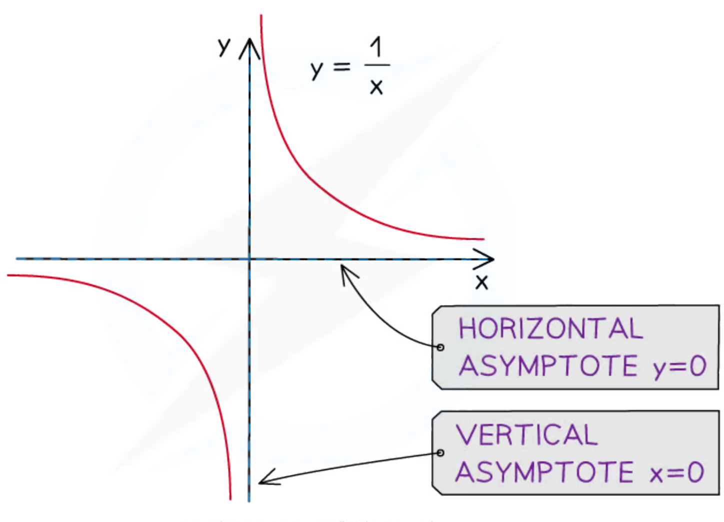


- You must also be able to recognise the three basic **trigonometric graphs**, covered in the Trigonometry section

Your notes

Where are the asymptotes on reciprocal graphs?

- An **asymptote** is a line on a graph that a curve becomes closer to but never touches
  - These may be horizontal or vertical
- The **reciprocal** graph, $y = \frac{a}{x}$ (where a is a constant)
  - **does not have a *y*-intercept**
  - **and does not have any roots**
- This graph has **two asymptotes**
  - A **horizontal** asymptote at the *x*-axis: $y = 0$
    - This is the **limiting value** when the value of *x* gets very large (or very negative)
  - A **vertical** asymptote at the *y*-axis: $x = 0$
    - This is the value that causes the **denominator to be zero**



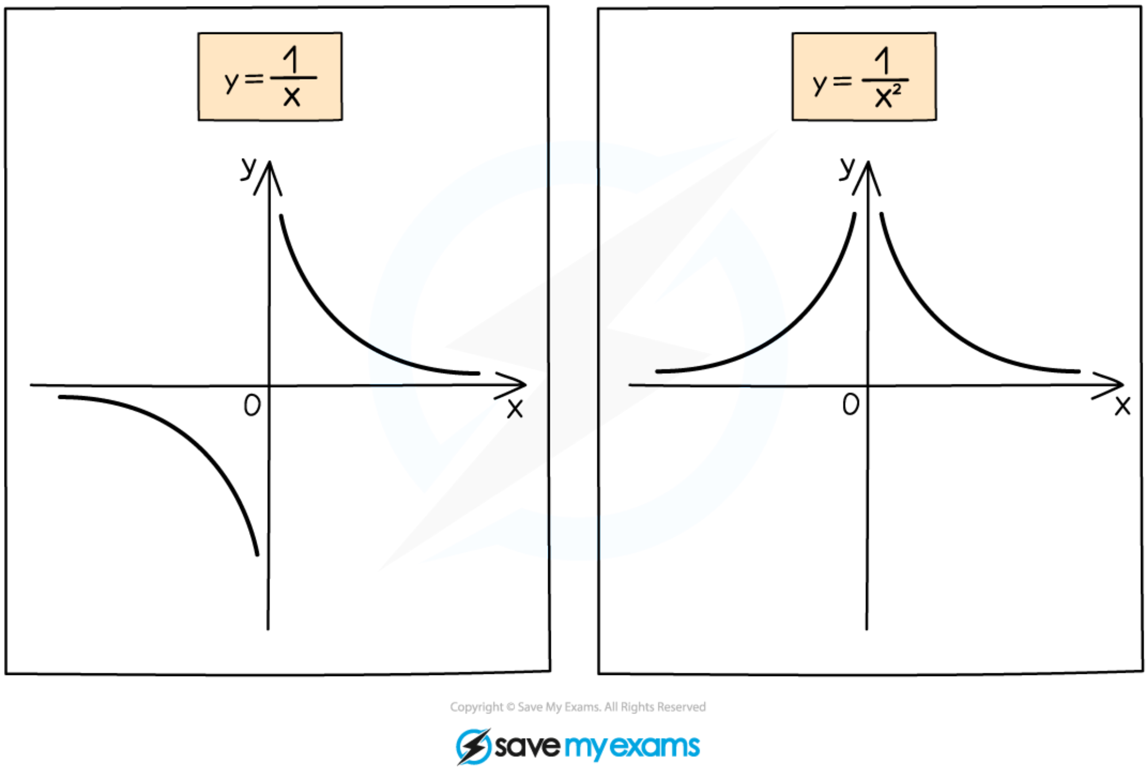
- The reciprocal graph, $y = \frac{a}{x} + b$ (where a and b are both constants)

- is the **same shape** as $y = \frac{a}{x}$
- but is **shifted upwards** by b units
  - $y = \frac{a}{x} - 3$ would be $y = \frac{a}{x}$ shifted **down** by 3 units
- This means the **horizontal asymptote** also **shifts up** by b units
  - The **vertical asymptote** remains on the ***y*-axis**

- The graph of $y = \frac{1}{x^2}$ is similar to $y = \frac{1}{x}$ but has two key differences
  - $y = \frac{1}{x^2}$ is **steeper** than $y = \frac{1}{x}$
  - $y = \frac{1}{x^2}$ is **always positive**, even when x is negative

Your notes

Your notes



How do I draw exponential growth and decay?

- The equation $y = k^x$ represents **exponential growth** when $k > 1$
  - $y = k^x$ represents **exponential decay** when $0 < k < 1$
    - k is positive but **less than 1**
- Both of these graphs:
  - have a **horizontal asymptote** at $y = 0$
  - **do not** have a **vertical asymptote**
  - have a y-intercept of $(0,\ 1)$
- The graph of $y = ak^x + b$ is a similar shape to $y = k^x$, but there are some differences
  - It is first **stretched** vertically by a

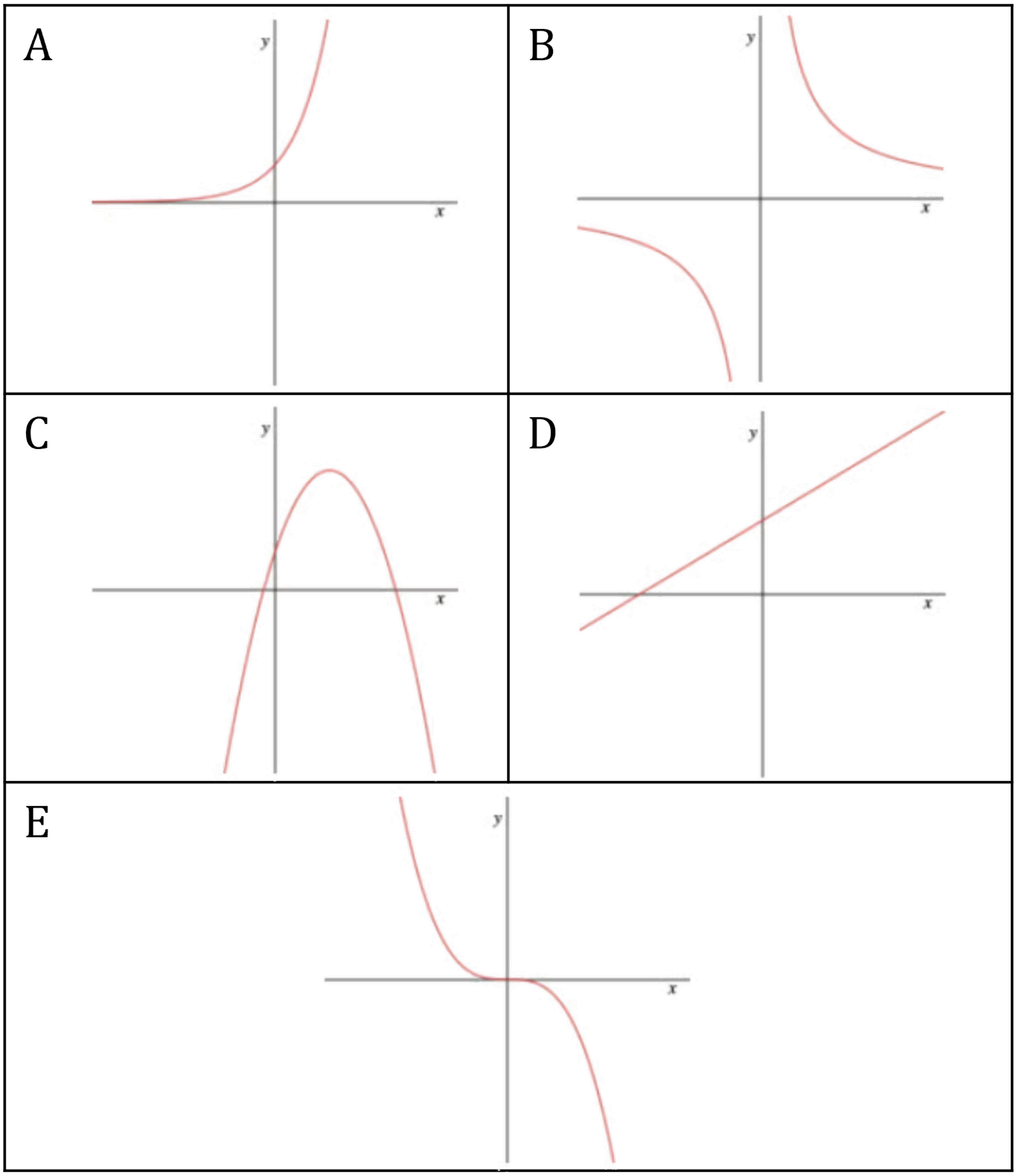
Your notes

- It is then **shifted** b units upwards
  - Therefore it has a **horizontal asymptote** at $y = b$
  - and a y-intercept of $(0,\ a + b)$
- For example, a population may be modelled as $y = 400 \times \left(\frac{1}{2}\right)^x + 100$, where y is the population and x represents time
  - This is an exponential decay as $0 < k < 1$
  - The initial population (when $x = 0$) will be 400 + 100 = 500
    - The y-intercept is (0, 500)
  - Over a long period of time (large x-value) the population will settle to 100
    - The asymptote is at $y = 100$
- **Exponential decay** can also be identified by a **negative power** using **index laws**
  - $\left(\frac{1}{2}\right)^x = (2^{-1})^x = 2^{-x}$ so $y = 400 \times 2^{-x} + 100$ is the model above
  - This has the form $y = k^{-x}$ where $k > 1$

Worked Example

Match the graphs to the equations.

Your notes



(1) $y = 0.6x + 2$, (2) $y = 3^x$, (3) $y = -0.7x^3$, (4) $y = \frac{4}{x}$, (5) $y = -x^2 + 3x + 2$

Starting with the equations,

(1) is a linear equation ($y = mx + c$) so matches the only straight line, graph **D**

(2) is an exponential equation with a positive coefficient so matches graph **A**

(3) is a cubic equation with a negative coefficient so matches graph **E**

(4) is a reciprocal equation with a positive coefficient so matches graph **B**

(5) is a quadratic equation with a negative coefficient so matches graph C

Graph A → Equation 2

Graph B → Equation 4

Graph C → Equation 5

Graph D → Equation 1

Graph E → Equation 3

Your notes

Drawing Graphs from Tables

Your notes

Drawing Graphs Using a Table

How do I draw a graph using a table of values?

- To create a **table of values**
  - **substitute** different ***x*-values** into the **equation**
    - This gives the ***y*-values**
- To plot the points
  - use the x and y-values to mark **crosses** on the grid at the **coordinates (*x* , *y*)**
    - Each point is expected to be plotted to **an accuracy within half of the smallest square** on the grid
- **Draw** a **single smooth freehand** curve
  - Go through **all** the plotted **points**
  - Make it the **shape** you would **expect**
    - For example, **quadratic** curves have a vertical line of **symmetry**
  - Do **not** use a ruler for curves!

Which numbers should I be careful with?

- For **quadratic** graphs, be careful substituting in **negative** numbers
  - Always put **brackets** around negative values and use **BIDMAS**
    - For example, substitute $x = -3$ into $y = -x^2 + 8x$
    - This becomes $y = -(-3)^2 + 8(-3)$
    - which simplifies to $-9 - 24$
    - so $y = -33$
- For **reciprocal** graphs like $y = \frac{1}{x}$, or $y = \frac{1}{x^2}$, do **not include x = 0**
  - You **cannot divide** by **zero**
    - You get an **error** on your **calculator**

Your notes

- There is **no value** at $x = 0$
  - The L-shaped branches **can't cross** the ***y*-axis**
  - There will be a **vertical asymptote** at $x = 0$
- An example is given below with $y = \frac{1}{x}$

x	-3	-2	-1	0	1	2	3
y	$-\frac{1}{3}$	$-\frac{1}{2}$	-1	No value	1	$\frac{1}{2}$	$\frac{1}{3}$

- You should also be careful when there is a **combination of different types of function**
  - E.g. $y = x^2 - \frac{1}{x} + 4$ has a quadratic term and a reciprocal term
  - This makes it **harder to know the shape** of the graph
  - But you can still **use a table of values** to plot them
  - Just be aware of points like $x = 0$ as described above, where there will be no value

How do I use the table function on my calculator?

- **Calculators** can create **tables of values** for you
- Find the **table** function
  - Type in the **graph** equation (called the **function**, f(x))
    - Use the alpha button then *X* or *x*
    - Press = when finished
  - If you are asked for another function, g(x), press = to ignore it
- Enter the **start** value
  - The first *x*-value in the table
  - Press =
- Enter the **end** value
  - The last *x*-value in the table

- Enter the **step size**
  - How big the steps (gaps) are from one *x*-value to the next
  - Press =
- Then scroll up and down to see all the ***y*-values**

Your notes

Examiner Tips and Tricks

- If you find a point that doesn't seem to fit the shape of the curve, check your working!
- If any *y*-values are given in the question, check that your calculations agrees with them

Worked Example

(a) Complete the table of values for the graph of $y = 10 - 8x^2$.

x	-1.5	-1	-0.5	0	0.5	1	1.5
y		2					-8

Use the table function on your calculator for $\mathrm{f}(x) = 10 - 8x^2$
Start at -1.5, end at 1.5 and use steps of 0.5

Alternatively, substitute the *x*-values into the equation, for example *x* = -1.5

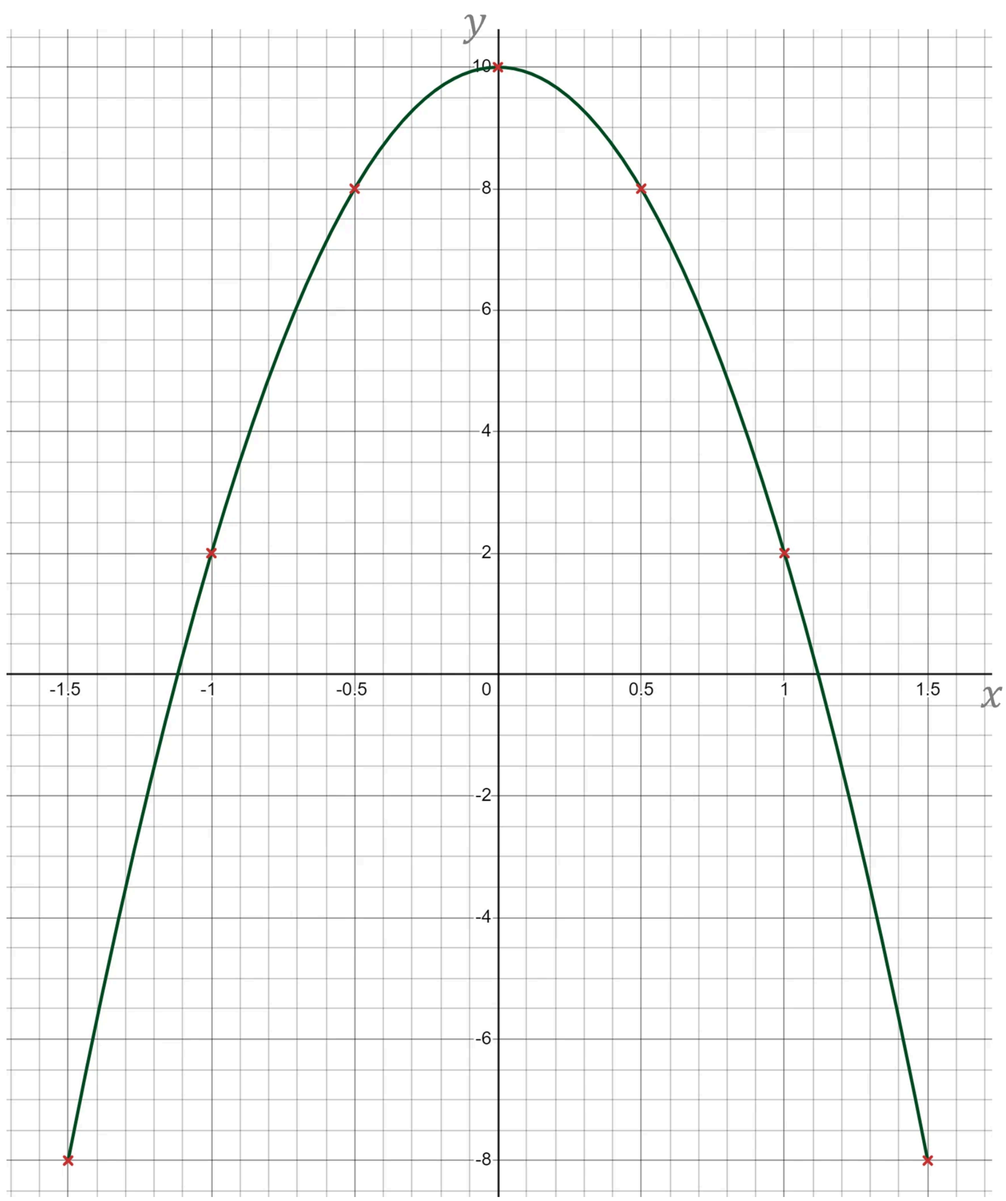
$$\begin{aligned} y &= 10 - 8(-1.5)^2 \\ &= 10 - 8 \times 2.25 \\ &= 10 - 18 \\ &= -8 \end{aligned}$$

x	-1.5	-1	-0.5	0	0.5	1	1.5
y	-8	2	8	10	8	2	-8

(b) Plot the graph of $y = 10 - 8x^2$ on the axes below, for values of x from -1.5 to 1.5.

Carefully plot the points from your table on to the grid
Note the different scales on the axes

Join the points with a smooth curve (do not use a ruler)



Your notes

(c) Write down the equation of the line of symmetry of the curve.

There is a vertical line of symmetry about the y-axis

The equation of the y-axis is $x = 0$

$x = 0$

Your notes

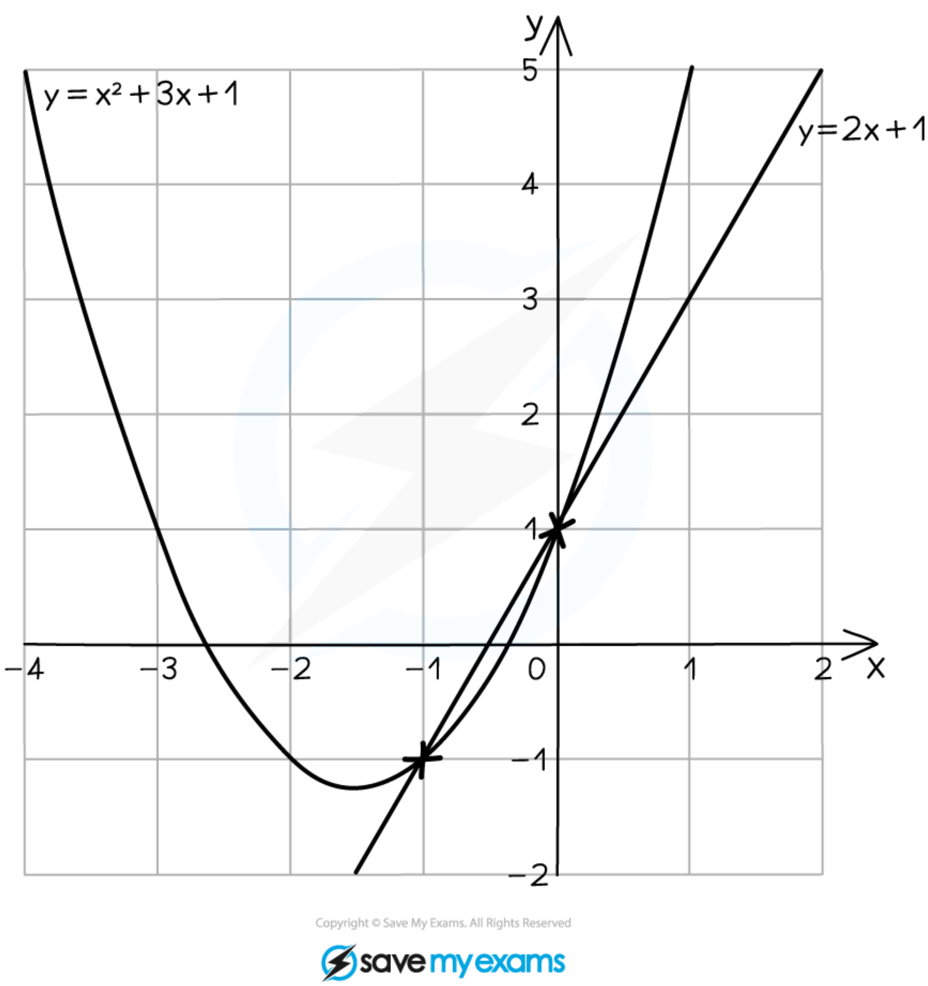
Your notes

Solving Equations from Graphs

Solving Equations Using Graphs

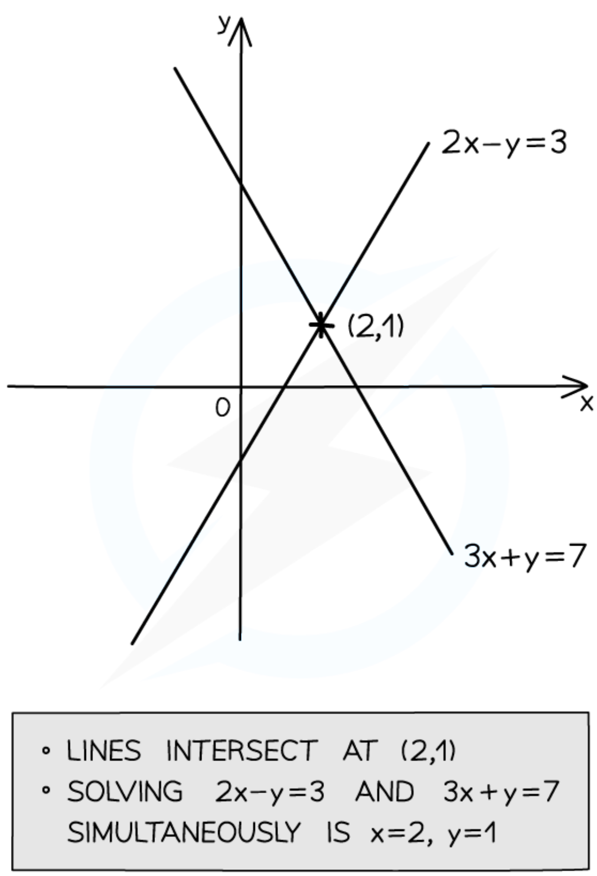
How do I find the coordinates of points of intersection?

- Plot **two graphs** on the **same** set of **axes**
  - The **points of intersection** are where the two lines **meet**
- For example, plot $y = x^2 + 3x + 1$ and $y = 2x + 1$ on the same axes
  - They meet twice, as shown
  - The **coordinates** of **intersection** are (-1, -1) and (0, 1)



How do I solve simultaneous equations graphically?

- The x and y **solutions** to **simultaneous equations** are the x and y **coordinates** of the **point of intersection**
- For example, to solve $2x - y = 3$ and $3x + y = 7$ simultaneously
  - **Rearrange** them into the form $\boldsymbol{y = mx + c}$
    - $y = 2x - 3$ and $y = -3x + 7$
  - Use a **table of values** to plot each line
  - Find the **point of intersection**, (2, 1)
  - The **solutions** are therefore $x = 2$ and $y = 1$



Your notes

How do I use graphs to solve equations?

- This is easiest explained through an example
- You can use the **graph** of $y = x^2 - 4x - 2$ to **solve** the following **equations**
  - $x^2 - 4x - 2 = 0$
    - The **solutions** are the two ***x*-intercepts**
    - This is where the curve cuts the *x*-axis (also called **roots**)
  - $x^2 - 4x - 2 = 5$
    - The **solutions** are the two ***x*-coordinates** where the curve **intersects** the **horizontal line** $y = 5$
  - $x^2 - 4x - 2 = x + 1$
    - The **solutions** are the two ***x*-coordinates** where the curve **intersects** the **straight line** $y = x + 1$
    - The straight line must be **plotted** on the same axes first
- To solve a **different** equation like $x^2 - 4x + 3 = 1$, if you are **already given** the graph of an equation, e.g. $y = x^2 - 4x - 2$
  - **add / subtract terms** to both sides to get "given graph = ..."
    - For example, subtract 5 from both sides
      - $x^2 - 4x - 2 = -4$
      - You can now draw on the horizontal line $y = -4$ and find the *x*-coordinates of the points of intersection

Examiner Tips and Tricks

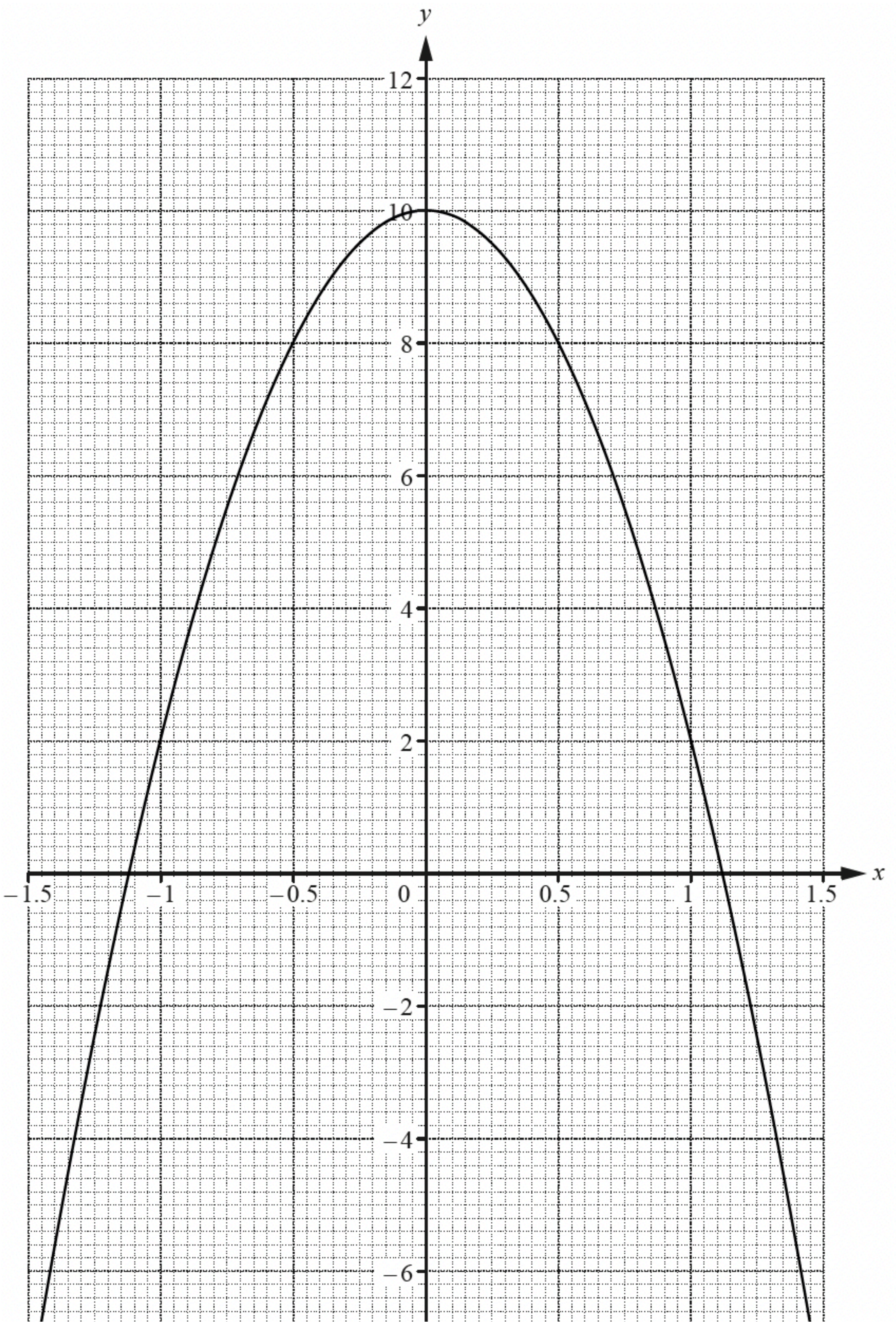
- When solving equations in *x*, only give ***x*-coordinates** as final answers
  - Include the *y*-coordinates if solving simultaneous equations

Your notes

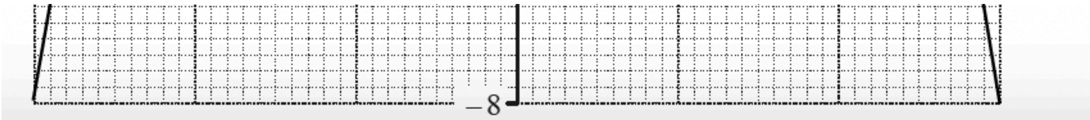
Worked Example

Use the graph of $y = 10 - 8x^2$ shown to estimate the solutions of each equation given below.

Your notes



Your notes



(a) $10 - 8x^2 = 0$

This equals zero, so the x-intercepts are the solutions
Read off the values where the curve cuts the x-axis
Use a suitable level of accuracy (no more than 2 decimal places from the scale of this graph)

-1.12 and 1.12

These are the two solutions to the equation

***x* = -1.12 and *x* = 1.12**

A range of solutions are accepted, such as "between 1.1 and 1.2"
Solutions must be ± of each other (due to the symmetry of quadratics)

(b) $10 - 8x^2 = 8$

This equals 8, so draw the horizontal line $y = 8$
Find the x-coordinates where this cuts the graph

-0.5 and 0.5

These are the two solutions to the original equation

***x* = -0.5 and *x* = 0.5**

The solutions here are exact

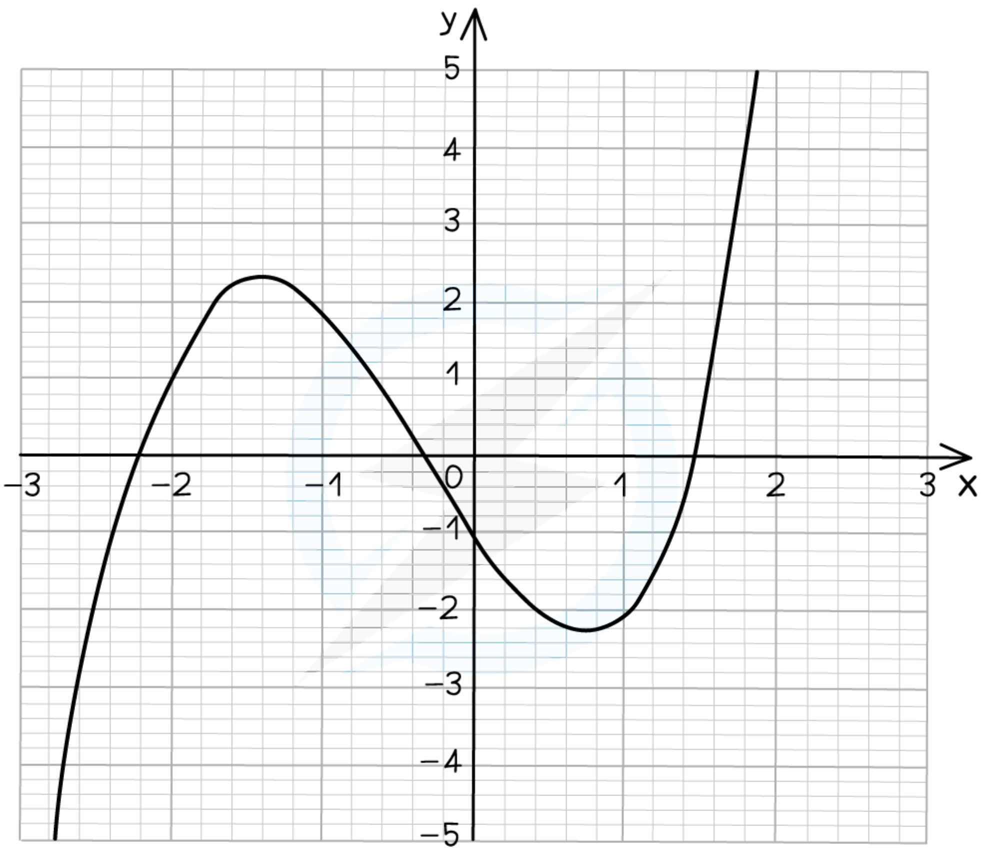
Worked Example

The graph of $y = x^3 + x^2 - 3x - 1$ is shown below.

Use the graph to estimate the solutions of the equation $x^3 + x^2 - 4x = 0$.

Give your answers to 1 decimal place.

Your notes



We are given a different equation to the one plotted so we must rearrange it to $\text{graph} = mx + c$, in this case $x^3 + x^2 - 3x - 1 = mx + c$

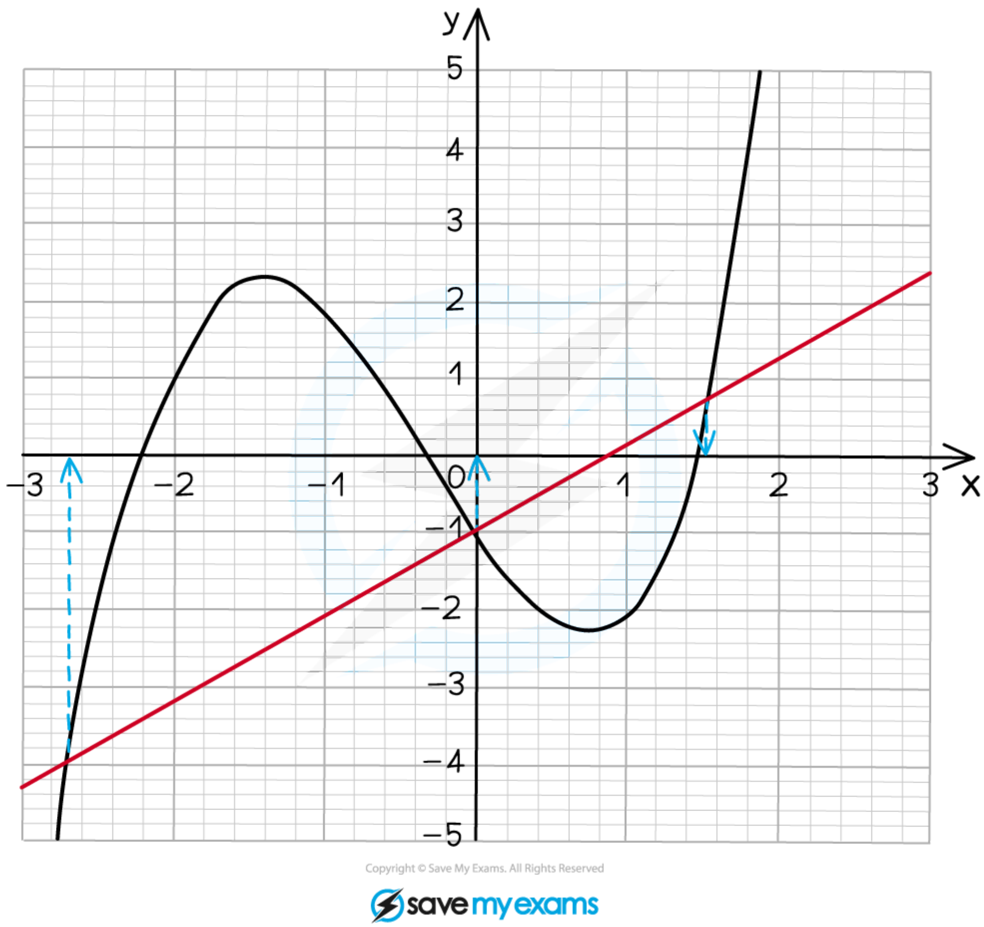
$$x^3 + x^2 - 4x = 0$$

$$+x - 1 \qquad\qquad +x - 1$$

$$x^3 + x^2 - 3x + 1 = x - 1$$

Now plot $y = x - 1$ on the same axes

Your notes



The solutions are the x-coordinates of where the curve and the straight line intersect

$$x = -2.6, \quad x = 0, \quad x = 1.6$$

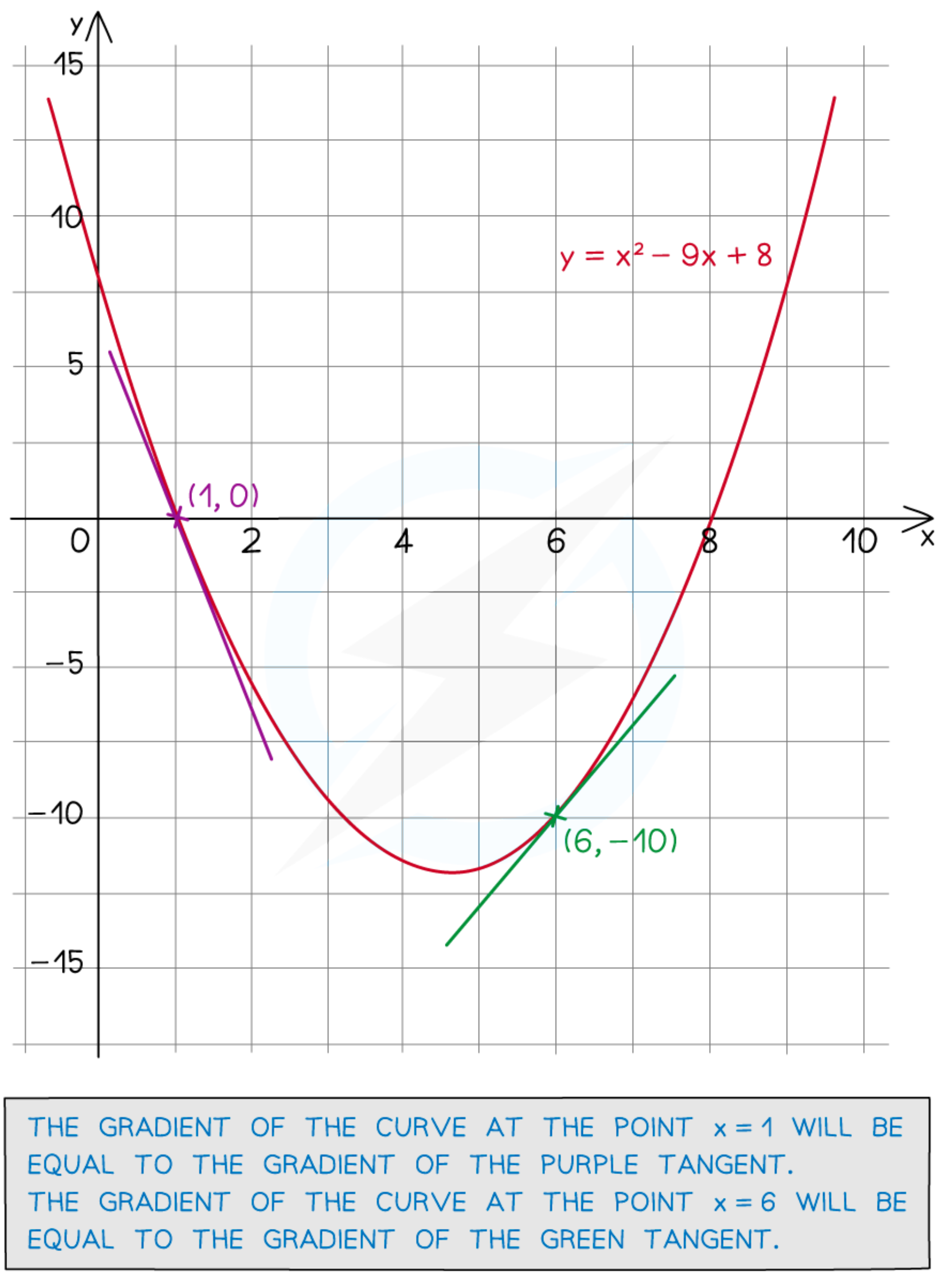
Your notes

Finding Gradients of Tangents

How are the gradients of graphs and tangents related?

- The **gradient of a graph at a point** is **equal** to the **gradient of the tangent** to the curve **at that point**
  - A tangent is a line that touches a curve, but does not cross it

Your notes

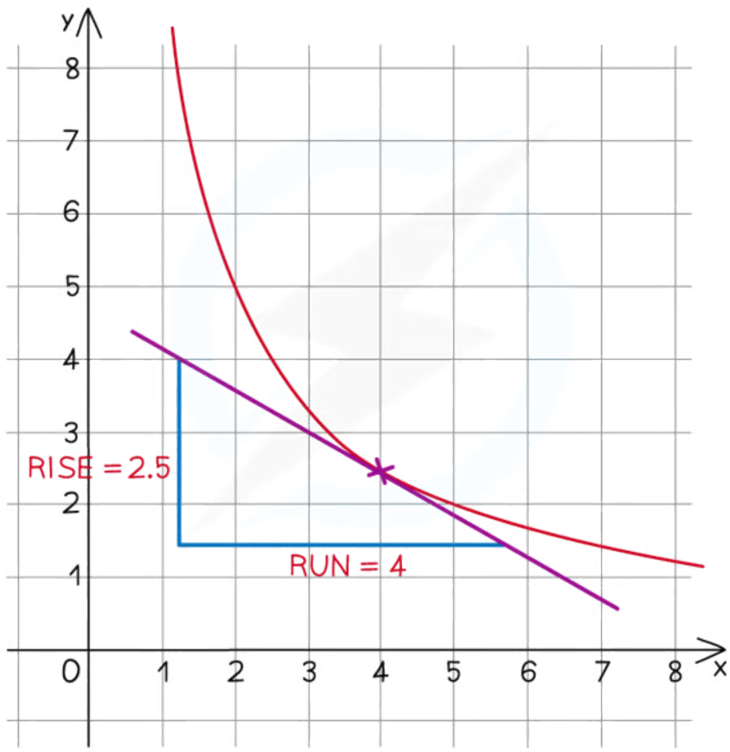


How do I estimate the gradient of a curve using a tangent?

- To find an **estimate** for the **gradient** of a curve **at a point**:
  - **Draw** a **tangent** to the curve at the **point**

Your notes

- Find the **gradient of the tangent** using
  - **Gradient = rise ÷ run**
  - or difference in y ÷ difference in x
  - In the example below, the **gradient of the tangent** at $x = 4$ would be $\frac{-2.5}{4} = -0.625$
  - Remember that the rise is **negative** if it is going down
  - This means the **gradient of the curve** at $x = 4$ is **also -0.625**



- It is an **estimate** because the **tangent** has been drawn by eye and is not exact
  - To find the exact gradient we would need to use **differentiation**

What does the gradient represent?

- The gradient represents the **rate of change** of y with x
  - I.e. For every increase in x by 1, how much does y increase?
- Consider the quantities used for the axes to determine the meaning of the gradient

- In a **distance-time graph**, the gradient is the rate of change of distance with time
  - This is the **speed**
- In a **speed-time graph**, the gradient is the rate of change of speed with time
  - This is the **acceleration**
- In a **graph of volume against radius**, e.g. as a balloon is inflated, the gradient is the rate of change of volume as the radius increases

Your notes

Examiner Tips and Tricks

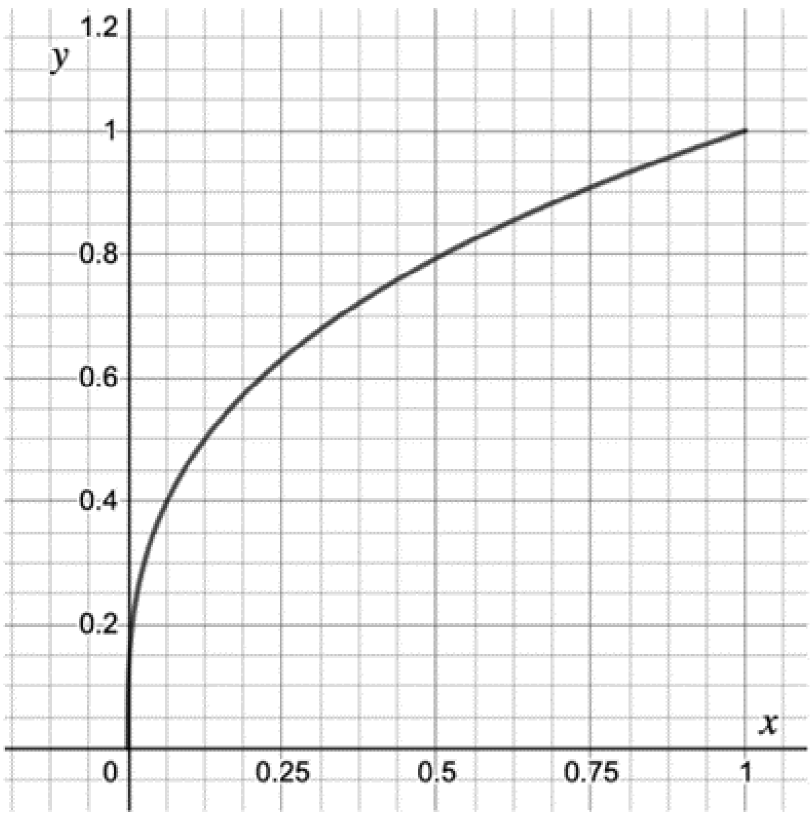
- When drawing a tangent by hand:
  - Use a ruler
  - Draw the line as long as you can
- When finding the gradient of the tangent:
  - Pick two points that are far away from one another
  - This will reduce the effect of any inaccuracy

Worked Example

The graph below shows $y = \sqrt[3]{x}$ for $0 \le x \le 1$.

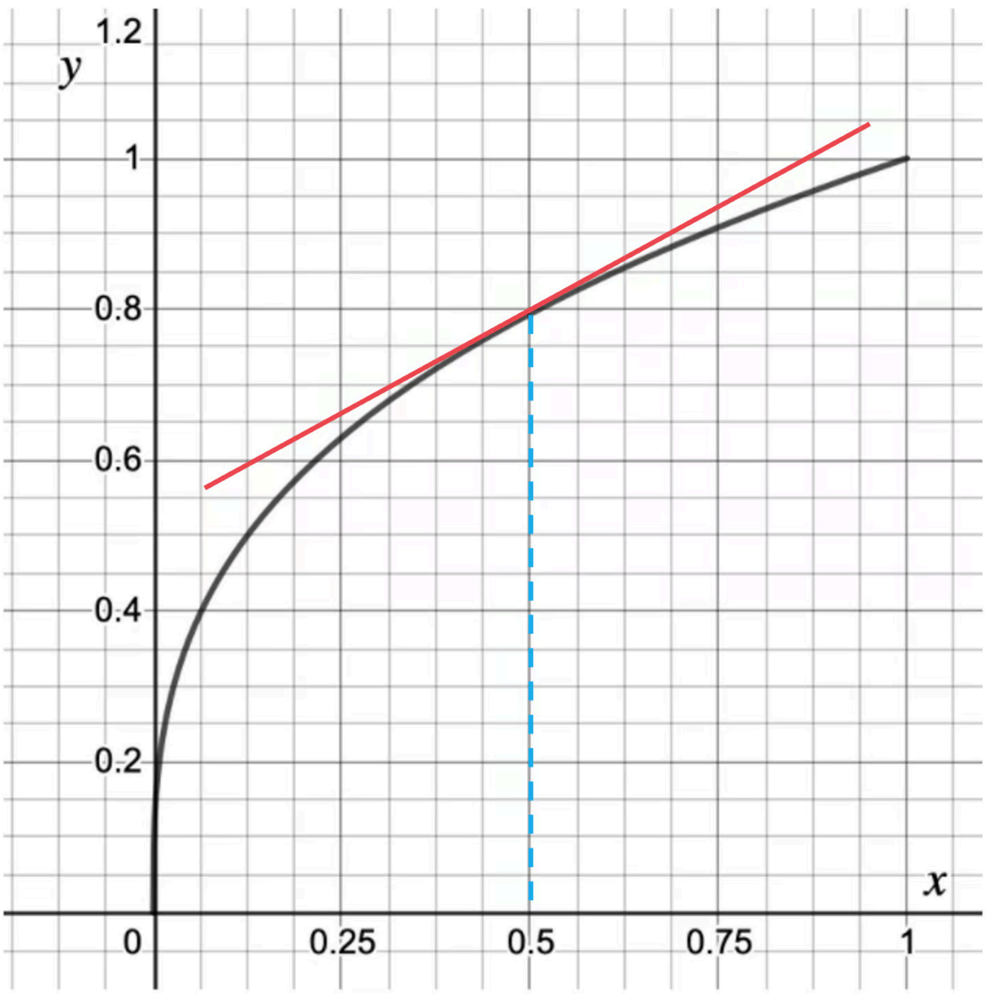
Find an estimate of the gradient of the curve at the point where $x = 0.5$.

Your notes



Draw a tangent to the curve at the point where $x = 0.5$

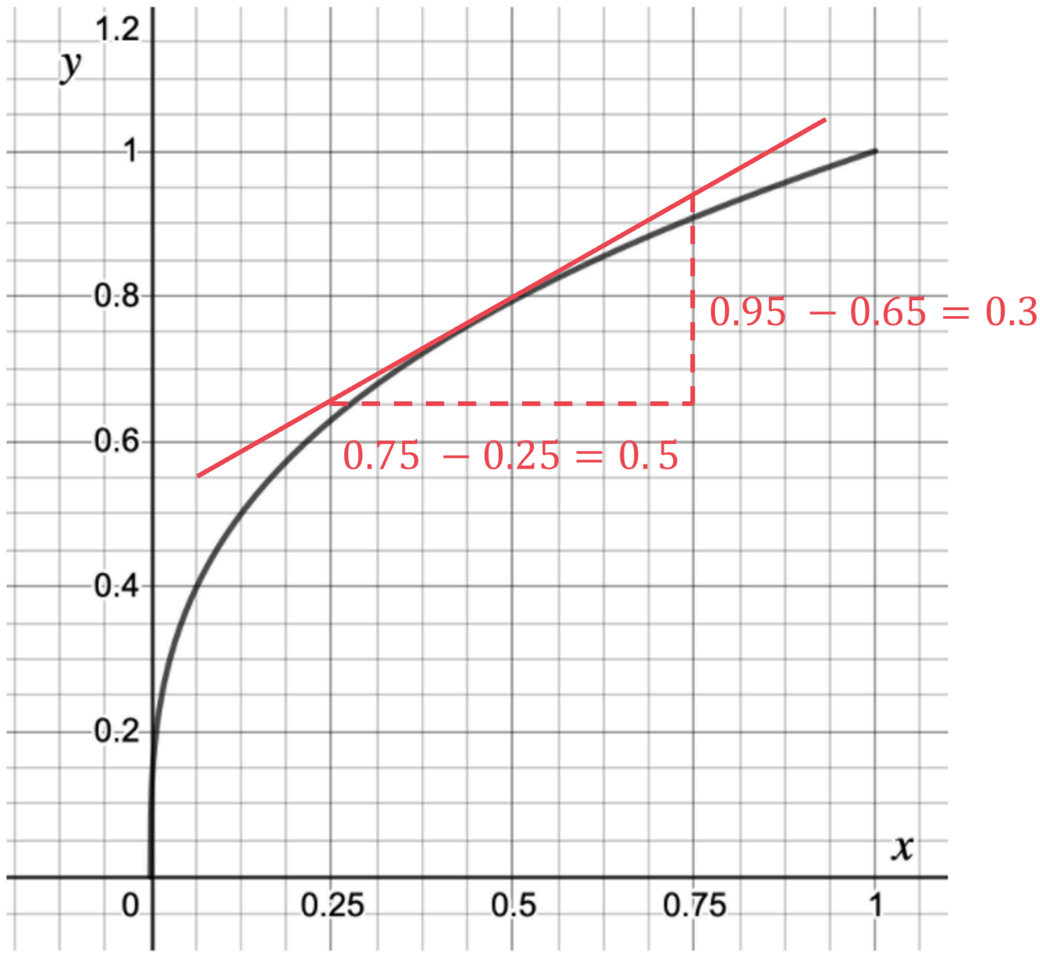
Your notes



Find suitable, easy to read coordinates as far apart as possible and draw a right-angled triangle between them

Find the difference in the *y* coordinates (rise) and the difference in the *x* coordinates (run).

Your notes



Find the gradient by dividing the difference in *y* (rise) by the difference in *x*

$$\text{Gradient } = \frac{0.3}{0.5} = \frac{3}{5} = 0.6$$

The gradient of the tangent at *x* = 0.5 is equal to the gradient of the curve at *x* = 0.5

Estimate of gradient = 0.6